JAIDEV EDUCATION SOCIETY'S

# J D COLLEGE OF ENGINEERING AND MANAGEMENT

KATOL ROAD, NAGPUR

Website: www.jdcoeng.ac.in E-mail: [illegible]

(An Autonomous Institute, with NAAC "A" Grade)

Affiliated to DBATU, RTMNU & MSBTE Mumbai

Department of Computer Science & Engineering

*"A place to learn; A chance to grow"*

Session: 2025 - 26 (Odd Semester)

MSE-II EXAMINATION

**Course: B.Tech. in CSE** | **Year/Semester: 3rd Semester (2nd Year)**

**Subject:Object Oriented Programming** | **Subject Code: CS3T003**

**Max Marks: 20** | **Date:01/10/2025** | **Duration: 1 Hr.**

**Instructions to the Students:**

1. **All the questions carry marks as indicate.**
2. **Assume suitable data whenever necessary.**
3. **Illustrate your answer whenever necessary with the help of neat sketches.**

**Students should be able to:**

CO1. Students will be able to understand the object oriented paradigm during development of application.

CO2. Students will be able to Study and apply the concept object initialization and destroy using constructors and destructors.

CO3. Students will be able to Develop application using the concept of inheritance and evaluate the usefulness.

CO4. Students will be able to classify & demonstrate the use of different data structures linked list, trees & graphs along with related algorithms.

CO5. Students will be able to create applications with the usage of different data structures using object oriented programming concept.

SECTION A

1. **All Questions are compulsory.**
2. **Answer in one word/sentence.**

| Q. No. | Questions | Level/CO Mapped | Marks |
|---|---|---|---|
| Q.1 | In the inheritance relationship class B extends A, B is called?<br>a) Base class b) Derived class c) Parent class d) None | L1/CO1 | 1 |
| Q.2 | Which constructor is called first in inheritance?<br>a) Derived class constructor b) Base class constructor<br>c) Child class constructor d) Both simultaneously | L1/CO1 | 1 |
| Q.3 | Runtime polymorphism in Java is achieved by:<br>a) Method overloading b) Method overriding<br>c) Constructor overloading d) Operator overloading | L1/CO2 | 1 |
| Q.4 | Static binding happens at:<br>a) Runtime b) Compile time c) Execution time d) After compilation | L1/CO2 | 1 |

SECTION B

1. **All Questions are compulsory.**
2. **Solve answer in brief (two, three Sentence).**

| | | | |
|---|---|---|---|
| Q.5 | Discuss is inheritance in Java, and why is it important?? | L2/CO1 | 2 |
| Q.6 | Describe constructor behavior in inheritance in Java. | L2/CO1 | 2 |
| Q.7 | Explain runtime polymorphism in Java. | L2/CO2 | 2 |
| Q.8 | Differentiate between static and dynamic binding in Java. | L4/CO2 | 2 |

SECTION C

1. **Solve any two questions**

| | | | |
|---|---|---|---|
| Q.9 | Write the Java program on Single and Multilevel Inheritance? | L5/CO1 | 4 |
| Q.10 | Describe how access modifiers affect inheritance in Java with a code example? | L4/CO1 | 4 |
| Q.11 | Explain function overloading with a Java program demonstrating method overloading for adding different types of numbers? | L5/CO2 | 4 |

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2025-2026(Odd Semester)

## MSE-2 EXAMINATION

**Course: B.Tech in CSE/AI** **Year/Semester: 3rd Semester (2nd Year)**

**Subject:** Innovation, Startup & Entrepreneurship **Subject Code:**

**Max Marks: 20** **Date:04/10/2025** **Duration: - 1 Hr.**

**Instructions to the Students:**

1. All the questions carry marks as indicated.
2. Assume suitable data whenever necessary.
3. Illustrate your answer whenever necessary with the help of neat sketches.

**Students should be able to:**

CO 1: Understand innovation processes and apply creativity techniques to generate viable ideas

CO 2: Analyze entrepreneurial fundamentals and demonstrate startup planning through practical exercises

CO 3: Evaluate startup ecosystem elements and create comprehensive business presentations

## SECTION A

1. **All Questions are compulsory.**
2. **Answer in one word/sentence.**

| Q. No. | Questions | Level/CO Mapped | Marks |
|---|---|---|---|
| Q.1 | Which of the following best describes an incubator?<br>a) Provides seed funding for equity<br>b) Offers mentorship, workspace, and early-stage support.<br>c) Organizes demo days for investors<br>d) Focuses on global expansion | L01/CO1 | 1 |
| Q.2 | Angel investors typically:<br>a) Invest institutional funds<br>b) Invest personal money in early-stage start-ups.<br>c) Focus only on later-stage companies<br>d) Avoid equity-based investment. | L01/CO1 | 1 |
| Q.3 | Which revenue model generates predictable, recurring income?<br>a) Asset Sale<br>b) Subscription<br>c) Brokerage Fee<br>d) One-time License | L02/CO2 | 1 |
| Q.4 | Economies of scope arise when:<br>a) Costs per unit fall with increased production<br>b) Costs decrease by producing a variety of products<br>c) Fixed costs spread over more units<br>d) Operations are automated | L01/CO2 | 1 |

## SECTION B

2. **Solve answer in brief (two, three Sentence).**

| | | | |
|---|---|---|---|
| Q.5 | State two advantages and two disadvantages of bootstrapping. | L01/CO2 | 2 |
| Q.6 | Explain the role of angel investors in supporting startups. | L02/CO3 | 2 |
| Q.7 | Differentiate between fixed costs and variable costs with examples. | L02/CO2 | 2 |
| Q.8 | What is a Minimum Viable Product (MVP) in Lean Startup methodology? | L01/CO2 | 2 |

## SECTION C

1. **Solve any two questions.**

| | | | |
|---|---|---|---|
| Q.9 | List and explain different types of company structures available in India. | L03/CO1 | 4 |
| Q.10 | Create a Business Model Canvas for an online grocery delivery start-up. | L03/CO1 | 4 |
| Q.11 | "Incubators and accelerators reduce the failure rate of startups." Justify the statement with examples. | L06/CO3 | 4 |

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Session: 2025-26

# MSE-2 EXAMINATION

**Course:** B. Tech. in Computer Science & Engineering (B) **Semester/ Year:** III Semester/ 2nd Year

**Subject:** Operating System **Subject Code:** CS3T001

**Max Marks:** 20 **Date:** 29/09/2025 **Duration:-** 1 Hr.

**Instructions to the Students:**
1. **All the questions carry marks as indicate.**
2. **Assume suitable data whenever necessary.**
3. **Illustrate your answer whenever necessary with the help of neat sketches.**

**Students should be able:**
1. To understand the basics of how operating systems work.
2. To explain how processes and CPU scheduling function in an operating system.
3. To solve common process synchronization problems.
4. To describe memory management concepts, including virtual memory.
5. To Comprehend disk management and the role of file systems in an operating system.

## SECTION A

1. **All Questions are compulsory.**
2. **Answer in one word/sentence.**

| Q. No. | Questions | Level/CO Mapped | Marks |
|---|---|---|---|
| Q.1 | Which of the following is *not* Deadlock Characterization?<br>A) Linux<br>B) Hold & Wait<br>C) No Preemption<br>D) Circular Wait | L1/CO3 | 1 |
| Q.2 | Which of the following is type of Virtual Memory?<br>A) Paging<br>B) Segmentation<br>C) Paging & Segmentation<br>D) None | L2/CO4 | 1 |
| Q.3 | The Banker's algorithm is used for:<br>A) Deadlock prevention<br>B) Deadlock avoidance<br>C) Deadlock detection<br>D) Recovery | L2/CO3 | 1 |
| Q.4 | Which of the following is Solution for Critical Section Problem?<br>A) Mutual Exclusion<br>B) Progress<br>C) Bounded Waiting<br>D) All of Above | L2/CO3 | 1 |

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Session: 2025-26

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## SECTION B

1. **All Questions are compulsory.**
2. **Solve answer in brief (two, three Sentence)**

| | | | |
|---|---|---|---|
| Q.5 | Write short note on Producer-Consumer Problem. | L3/CO3 | 2 |
| Q.6 | Define Synchronization in the context of operating systems with its types. | L1/CO3 | 2 |
| Q.7 | Describe the Dining Philosophers Problem with its significance in process synchronization. | L2/CO3 | 2 |
| Q.8 | Discuss the various methods used for handling deadlocks in operating systems. | L2/CO3 | 2 |

## SECTION C

1. **Solve any two questions.**

| | | | |
|---|---|---|---|
| Q.9 | Distinguish between internal fragmentation and external fragmentation with suitable examples. | L4/CO4 | 4 |
| Q.10 | Write short note on i) Peterson Solution ii) Semaphores | L3/CO4 | 4 |
| Q.11 | Explain how the Memory Management Unit (MMU) translates logical addresses to physical addresses step-by-step. | L3/CO4 | 4 |

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# MSE-II EXAMINATION

**Course:** B. Tech. in Computer Science & Engineering **Semester/ Year:** III Semester/ 2nd Year

**Subject:** Analytical Skills and Ethical Application **Subject Code:** CS3V001

**Max Marks: 20** **Date: 06/10/2025** **Duration:- 1 Hr.**

**Instructions to the Students:**
1. All the questions carry marks as indicate.
2. Assume suitable data whenever necessary.
3. Illustrate your answer whenever necessary with the help of neat sketches.

**Students should be able to:**
1. Illustrate the use of number system rules to solve basic simplification problems.
2. To analyze and solve real-life and mathematical problems involving percentages, including population variations and percentage-based changes.
3. To apply and evaluate profit and loss concepts to solve problems involving cost price, selling price, marked price, and discounts effectively.
4. To apply formulas for simple and compound interest to solve practical and mixed-interest problems.
5. To analyze mathematical concepts related to ratios, proportions, partnerships, and age problems to solve both theoretical and practical scenarios.

| | | Level/CO | Marks |
|---|---|---|---|
| | **Objective type questions (Compulsory Question)** | | |
| Q.1 | A vendor buys oranges at Rs 26 per dozen and sells them at 5 for Rs 13. Find his gain percent.<br>a. 4% b. 5% c. 10% d. 20% | L3/CO2 | 1 |
| Q.2 | By selling an article for Rs 240, a man makes a profit of 20%. What would his profit percent be if he sold the article for Rs 275?<br>a. 35.5% b. 37.5% c. 30% d. 32.5% | L2/CO1 | 1 |
| Q.3 | Rama purchased a bike for Rs. 54000. He sold it at a loss of 8 percent. With that money he again purchased another bike and sold it at a profit of 10 percent. What is his overall loss or profit?<br>a. Loss of Rs. 657 b. Profit of Rs. 657 c. Loss of Rs. 648 d. Profit of Rs. 648 | L3/CO1 | 1 |
| Q.4 | A sells a car to B at 10% loss. If B sells it for Rs. 54000 and gains 20%, the cost price of the car for A was?<br>a. 25000 b. 50000 c. 37500 d. 60000 | L2/CO1 | 1 |
| Q.5 | If selling price of 40 articles is equal to cost price of 50 articles, the loss or gain percent is?<br>a.25% loss b. 20% loss c.20% profit d.25% profit | L2/CO1 | 1 |
| Q.6 | The ratio of cost price and selling price is 4:5. The profit percent is?<br>a. 10% b. 20% c. 25% d. 30% | L3/CO1 | 1 |
| Q.7 | A man bought an old typewriter for Rs.1200 and spent Rs. 200 on its repair. He sold it for Rs.1680. His profit per cent is: | L3/CO1 | 1 |

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a. 20% b. 10% c. 8% d. 16%

Q.8 A milkman bought 70 litres of milk for Rs.630 and added 5 litres of water. If he sells it at Rs. 9.00 per litre, his profit percentage is: L2/CO1 1

a. 8 ⅕ % b. 7% c. 8 ⅔% d. 7 ⅐%

Q.9 A sum of money placed at compound interest doubles itself in 4 years. In how many years will it amount to 8 times? L2/CO1 1

a. 9 years b. 8 years c. 27 years d. 12 years

Q.10 The cost price of 20 article is same as the selling price of 'X' article. If the profit is 25% , then the value of 'X'. L3/CO1 1

a. 20 b. 22 c. 16 d. 18

Q.11 A sum of Rs.6000 is deposited into ICICI bank for 4 years. If the bank provides 6%, then what is the amount after the maturity period? L4/CO1 1

a. 7440 b. 6440 c. 1440 d. 2440

Q.12 An article was sold for Rs. 5220 at a loss of 42% of the cost price. What will be the selling price of the article for a profit of 42%? L2/CO1 1

a. 12580 b. 17280 c. 12780 d. None of these

Q.13 At what rate of simple interest, will a sum of money double itself in 4 years? L3/CO2 1

a. 10% b. 20% c. 25% d. 15%

Q.14 A sum fetched a total simple interest of Rs. 4016.25 at the rate of 9 p.c.p.a. in 5 years. What is the sum? L4/CO2 1

a. ₹4462.50 b. ₹8032.50 c. ₹8900 d. ₹8925

Q.15 On the eve of Gandhi Jayanti a saree is sold for Rs. 720 after allowing 20% discount. What is its marked price? L4/CO2 1

a. ₹740 b. ₹900 c. ₹864 d. ₹950

Q.16 A jacket is marked at Rs. 80 and is offered with two successive discounts of 10% and 5%. What is the final selling price? L3/CO2 1

a. 68.40 b. 74.00 c. 3000 d. 3200

Q.17 A book with a list price of Rs.50 is sold for Rs.40. What is the discount percentage? L4/CO2 1

a. 10% b. 20% c. 30% d. 40%

Q.18 Find the simple interest on Rs. 68,000 at 16 ⅔% per annum for 6 months? L3/CO2 1

a. 5666.67 b. 5888.7 c. 5662 d. None of these

Q.19 Find the compound interest on Rs. 15,625 for 6 months at 16% per annum compounded quarterly? L2/CO1 1

a.1575 b. 1275 c. 1475 d. 1375

Q.20 What will be the difference between single discount of 30% on Rs. 2000 and 2 successive discounts of 15% and 15%? L3/CO1 1

Rs 0 b. Rs. 10 c. Rs 45 d. Rs. 150

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## Department of CSE/AI

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Session: 2025-26 (Odd Semester)

MSE-2 EXAMINATION

**Course: B. Tech. Subject:** SADM
**Subject Code:** CS3M001
**Year/Semester:** III[rd] Semester (2[nd] Year)
**Max Marks: 20** **Date :- 03/10/2025** **Duration:- 1 Hr**

**Instructions to the Students:**

1. **All the questions carry marks as indicate.**
2. **Assume suitable data whenever necessary.**
3. **Illustrate your answer whenever necessary with the help of neat sketches.**

CO 1: Define fundamental concepts of linear algebra, numerical methods, probability, and Group Theory.

CO 2 : Illustrate the concepts of matrices, numerical techniques, Group Theory methods and probability distributions with relevant examples.

CO 3: Apply linear algebra techniques, numerical methods, and Probability and Group Theory to solve engineering mathematical problems.

CO 4: Analyze the concept of probability distributions, numerical technique, Group Theory, and linear algebra technique.

CO 5: Evaluate the mathematical problem by using , Eigen values and Eigen vectors, Sylvester's theorem, Bisection method, Newton- Raphson method and their convergences. Solution of system of linear equations by Gauss Seidel method, Gauss elimination method, Crout's method. Mathematical Expectation, probability and Group Theory.

CO 6: Develop Mathematical Models based on linear algebra and probability distributions & Group Theory to solve real-life and engineering-related Problems.

## SECTION A

**All Questions are compulsory.**
**Answer in one word/sentence.**

| Q. No. | Questions | Level/CO Mapped | Marks |
|---|---|---|---|
| Q.1 | Find the Variance of $X$, where $E(X^2) = 10$ and $E(X) = 2$<br>(a) 6 (b) 1 (c) -6 (d) 14 | L1/CO1 | 1 |
| Q.2 | Tell the Formula for moment generating function in Discrete Case?<br>(a) $\sum e^{tx} f(x)$ (b) $\sum e^{-t} f(x)$ (c) $\sum e^{t} f(x)$ (d) $\sum e^{-tx} f(x)$ | L1/CO1 | 1 |
| Q.3 | Choose the correct option for Independent Random Variable<br>(a) $f_1(x).f_2(y)=f(x,y)$ (b) $f_1(x).f_2(y) \neq f(x,y)$<br>(c) $f_1(x).f_2(y)=0$ (d) All of these | L1/CO1 | 1 |

Q.4 Tell formula for $E(X) = \int_{-\infty}^{\infty} x\, f(x)\, dx$. L1/CO1 1

(a) Continuous Variable (b) Discrete Variable

(c) Density function (d) None of these

## SECTION B

**All Questions are compulsory.**

**Solve answer in brief (two, three Sentence)**

Q.5 Construct a Random Variable X as:- L2/CO2 2

| X | -2 | 3 | 1 |
|---|---|---|---|
| f(x) | 1/3 | 1/2 | 1/6 |

i) E(X) ii) $E(X^2+5X)$

Q.6 Illustrate the given Density Function L2/CO2 2

$$f(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0 \quad , & otherwise \end{cases}$$ ; find the value of E(X)

Q.7 Compare X and Y be a Random Variable having joint density function L2/CO2 2

$$f(x,y) = \begin{cases} Cxy & ; 0 \leq x \leq 1; 0 \leq y \leq x \\ 0 & ; \quad otherwise \end{cases}$$ ; find the value of C

Q.8 Construct X be a random variable with probability distribution as follows: L2/CO2 2

| X | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| f(x) | 1/3 | 1/2 | 0 | 1/6 |

Find the Expected value of $(X-1)^2$

## SECTION C

**Solve any two questions.**

Q.9 Analyze the value of X and Y be Random Variable having Joint Probability Function L4/CO4 4

$$f(X,Y) = \begin{cases} Cxy & ; x = 1,2,3\,; y = 1,2,3 \\ 0 & ; otherwise \end{cases}$$

1)Find the value of C.

2)Identify P(X>1) and P(X=3).

Q.10 Identify the given Density Function L4/CO4 4

$$f(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0 \quad , & otherwise \end{cases}$$

i) $E(X^3)$ ii) Var(X)

Q.11 Let X and Y be a Random Variable having joint density function L4/CO4 4

$$f(x,y) = \begin{cases} 8xy & ; 0 \leq x \leq 1; 0 \leq y \leq x \\ 0 & ; otherwise \end{cases}$$

i) E(X) and V(X)

ii) E(Y) and V(Y)

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## MSE-2 EXAMINATION

**Course: B tech in CSE** | **Year/Semester: 3rd Semester (2nd Year)**

**Subject: Data Structure Using Python (DSUP)** | **Subject Code: CS3T002**

**Max Marks: 20** | **Date: 30/9/2025** | **Duration: - 1 Hr.**

**Instructions to the Students:**

1. All the questions carry marks as indicate.
2. Assume suitable data whenever necessary.
3. Illustrate your answer whenever necessary with the help of neat sketches.

**Students should be able to:**

**CO 1: Students will be able to understand the basic concept of data structures, time complexity and analyse the various sorting and searching algorithms by using python**

**CO 2: Students will be able to implement dynamic data structures like singly, doubly and circular linked list.**

**CO 3: Students will be able to Apply the different linear data structures like stack and queue to various computing problems.**

**CO 4: Students will be able to Implement different types of trees and apply them to problem solution.**

**CO 5: Students will be able to Demonstrate the representation of graphs and their applications in real life problem**

## SECTION A

1. All Questions are compulsory.
2. Objective type question (Compulsory Question)

| Q. No. | Questions | Level/CO Mapped | Marks |
|---|---|---|---|
| Q.1 | Choose Which type of linked list allows traversal in both directions?<br>a) Singly Linked List b) Circular Linked List<br>c) Doubly Linked List d) Queue | L3/CO3 | 1 |
| Q.2 | Solve postfix expression 2,3,1,*,+,9,- evaluates to?<br>a) -30 b) 36 c) -8 d) 12 | L3/CO3 | 1 |
| Q.3 | Find Depth and Height of a node 10 in following Tree?<br> <br><br>a] 1,2 b]2,1 c] 3,2 d]2,3 | L1/CO4 | 1 |
| Q.4 | Choose the overflow condition in simple queue.<br>a] Real=N-1 b]Real= -1 c]real=front+size%capacity d]front=N-1 | L3/CO3 | 1 |

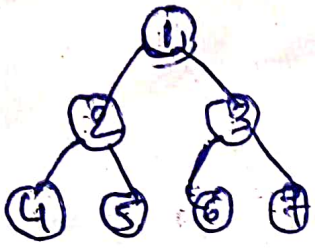
1. **All Questions are compulsory.**
2. **Solve answer in brief (two, three Sentence).**

Q.5 Convert the following Infix Expression to Prefix Expression L2/CO3

(A+B)*(C÷D) L2/CO2

Q.6 Discuss Quick Sort with Example.

Q.7 Find the Inorder, Preorder, Postorder Traversal of following Tree.




L1/CO4

Q.8 Write a Python Program for Doubly Linked List. L3/CO3

## SECTION C

1. **Solve any two questions.**

Q.9 ~~Describe~~ Singly Linked List with Insertion and Deletion operation. L2/CO3 4

Q.10 Explain Circular Queue and Perform Following Operation on Queue with capacity =4

1]Enqueue 30 2] Enqueue 22 3] Enqueue 13 4] Deque L2/CO3 4

5]Enqueue 55 6] Enqueue 42

Q.11 Draw Step by Step binary search tree using following data

45, 15, 79, 90, 10, 55, 12, 20, 50. L3/CO4 4